

# Chapter 1

## The Bretherton problem: An air finger propagates into a 2D channel

This tutorial discusses the solution of the classical Bretherton problem in which an air-finger is driven steadily into a 2D fluid-filled, rigid-walled channel.

The driver code listed below was originally developed as a test-bed for the paper

Hazel, A.L. & Heil, M. (2008) The influence of gravity on the steady propagation of a semi-infinite bubble into a flexible channel. *Physics of Fluids* 20, 092109 ([pdf reprint](#))

in which we studied the elastic-walled equivalent, paying particular attention to the effect of transverse gravity which plays quite an important role in this problem.

Compared to other tutorials, there is no detailed discussion of the driver code itself because the implementation is somewhat messy. However, given that the driver code is (of course!) very well documented you should be able to figure out what's going on once you've read through the discussion of the problem formulation and our brief comments on the [Implementation](#). [Get in touch](#) if you need more information.

### 1.1 The problem

The sketch below shows the problem setup: An (inviscid) air finger propagates at a steady speed  $U$  into a 2D rigid-walled, fluid-filled channel of half-width  $H_0^*$ . In the process it displaces some of the viscous fluid (of density  $\rho$ , viscosity  $\mu$  and surface tension  $\sigma^*$ ) and deposits a film of thickness  $h_0^*$  on the channel walls. [Note that, for the sake of simplicity, we ignore the effect of transverse gravity in this discussion; the driver code listed below allows the specification of a gravitational body force which will break the up-down symmetry of the solution.]

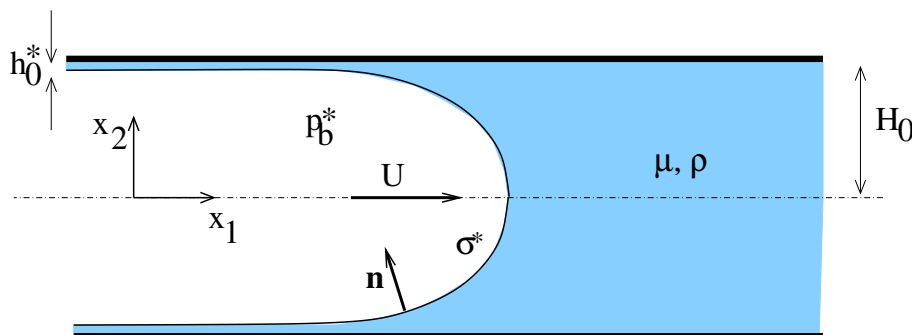


Figure 1.1 Sketch of the problem setup

We non-dimensionalise the velocities on  $U$ , the lengths on  $H_0^*$  and the pressure on the viscous scale,  $\mu U / H_0^*$  and solve the problem in a frame moving with the velocity of the finger. The flow is then governed by the non-dimensional Navier-Stokes equations

$$Re u_j \frac{\partial u_i}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

and the continuity equation

$$\frac{\partial u_i}{\partial x_i} = 0,$$

where the Reynolds number is defined as  $Re = U H_0 \rho / \mu$ . On the free fluid surface, whose outer unit normal we denote by  $\mathbf{n}$ , the fluid normal velocity vanishes,

$$\mathbf{u} \cdot \mathbf{n} = 0 \quad \text{on the air-liquid interface.}$$

We measure the fluid pressure  $p$  relative to the bubble pressure  $p_b$ . Then the dynamic boundary condition implies that

$$-pn_i + \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) n_j + \frac{1}{Ca} \kappa_f n_i = 0 \quad \text{on the air-liquid interface,}$$

where  $\kappa_f = \kappa_f^* H_0$  is the non-dimensional interface curvature and the capillary number  $Ca = U \mu / \sigma^*$  is a non-dimensional measure of the bubble velocity. The no-slip condition on the channel walls requires that  $\mathbf{u} = (-1, 0)$  at  $x_2 = \pm 1$ . Far behind the bubble tip, the fluid remains stationary on the walls. In the moving frame of reference, this corresponds to the plug flow profile  $\mathbf{u} = (-1, 0)$  as  $x_1 \rightarrow -\infty$ . The residual film thickness  $h_0$  has to be determined as part of the solution and it determines the volume flux through the system. The Poiseuille velocity profile far ahead of the bubble tip,

$$\mathbf{u} = \left( -1 + \frac{3}{2}(1 - h_0)(1 - x_2^2), 0 \right) \quad \text{as } x_1 \rightarrow \infty, \quad (1)$$

is then determined by overall continuity.

## 1.2 Some results

Here are some pretty pictures of the computed flow field,

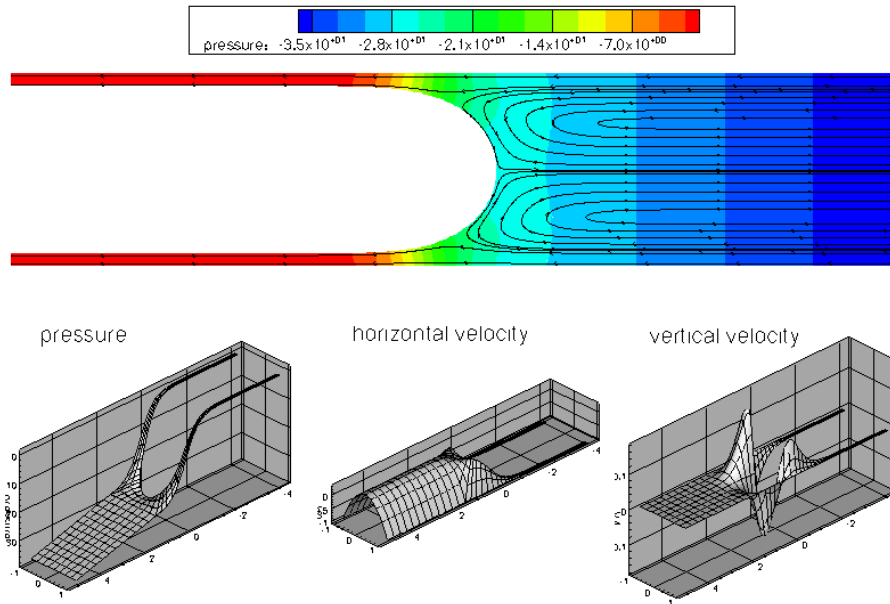


Figure 1.2 Velocity and pressure fields

and here is a comparison of the computational predictions for the bubble pressure and film thickness against Bretherton's famous asymptotic predictions (valid only for small capillary numbers!):

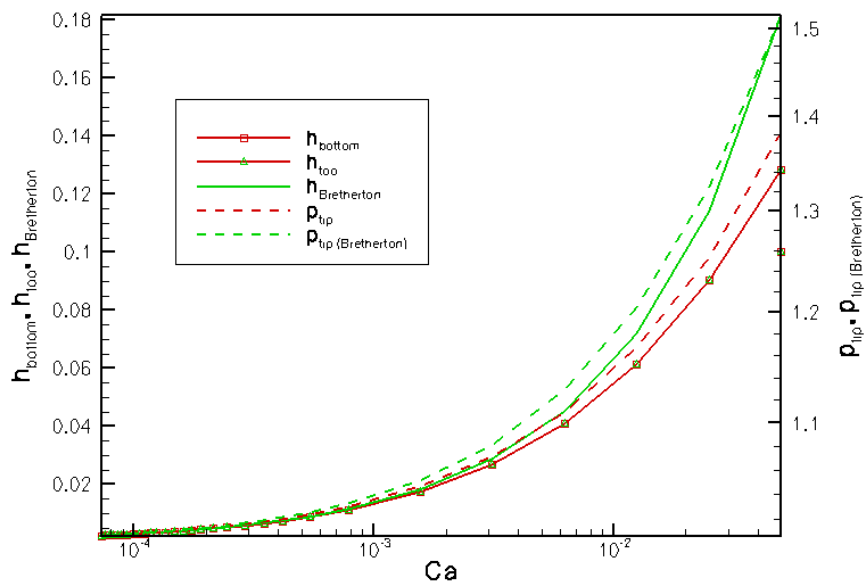


Figure 1.3 Bubble pressure and film thickness far behind the finger tip vs. capillary number; computed solution and Bretherton's predictions for small  $Ca$ .

## 1.3 Implementation

The discretisation of the problem is reasonably straightforward, the most (human-)time-consuming part being the creation of the spine mesh, discussed in [another tutorial](#). The real complication arises from the fact that the application of the "inflow condition" (1) at the right end of the domain – superficially a Dirichlet condition for the velocity – depends on the non-dimensional film thickness  $h_0$  at the left end of the domain. This introduces a non-local interaction between these degrees of freedom. We handle this by providing a templated wrapper class `BrethertonElement` (templated by the type of the "wrapped" fluid element) which allows the specification of the film thickness as external Data (i.e. Data whose values affect the element's residuals but are not determined by the element; see the discussion of oomph-lib's various types of elemental Data in [the "bottom up" discussion of the library's data structure](#)). Read the driver code listed below to see how it works!

## 1.4 The driver code

```
//LIC// =====
//LIC// This file forms part of oomph-lib, the object-oriented,
//LIC// multi-physics finite-element library, available
//LIC// at http://www.oomph-lib.org.
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//LIC// 02110-1301 USA.
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//LIC// The authors may be contacted at oomph-lib@maths.man.ac.uk.
//LIC//
//LIC// =====
// The 2D Bretherton problem
```

```

#include<algorithm>
// The oomphlib headers
#include "generic.h"
#include "navier_stokes.h"
#include "fluid_interface.h"

// The mesh
#include "meshes/bretherton_spine_mesh.h"

using namespace std;
using namespace oomph;

//=====
// Namespace for global parameters
//=====
namespace Global_Physical_Variables
{

    /// Reynolds number
    double Re;

    /// Womersley = Reynolds times Strouhal
    double ReSt;

    /// Product of Reynolds and Froude number
    double ReInvFr;

    /// Capillary number
    double Ca;

    /// Direction of gravity
    Vector<double> G(2);

    /// Pointer to film thickness at outflow on the lower wall
    double* H_lo_pt;

    /// Pointer to film thickness at outflow on the upper wall
    double* H_up_pt;

    /// Pointer to y-position at inflow on the lower wall
    double* Y_lo_pt;

    /// Pointer to y-position at inflow on the upper wall
    double* Y_up_pt;

    /// Set inflow velocity, based on spine heights at outflow
    /// and channel width at inflow
    void inflow(const Vector<double>& x, Vector<double>& veloc)
    {
#ifdef PARANOID
        std::ostringstream error_stream;
        bool throw_error=false;
        if (H_lo_pt==0)
        {
            error_stream << "You must set H_lo_pt\n";
            throw_error = true;
        }
        if (H_up_pt==0)
        {
            error_stream << "You must set H_up_pt\n";
            throw_error = true;
        }
        if (Y_lo_pt==0)
        {
            error_stream << "You must set Y_lo_pt\n";
            throw_error = true;
        }
        if (Y_up_pt==0)
        {
            error_stream << "You must set Y_up_pt\n";
            throw_error = true;
        }

        //If one of the errors has occurred
        if(throw_error)
        {
            throw OomphLibError(error_stream.str(),
                                OOMPH_CURRENT_FUNCTION,
                                OOMPH_EXCEPTION_LOCATION);
        }
#endif

        // Get average film thickness far ahead
        double h_av=0.5*(*H_lo_pt+*H_up_pt);

        // Get position of upper and lower wall at inflow

```

```

double y_up=*Y_up_pt;
double y_lo=*Y_lo_pt;

// Constant in velocity profile
double C =6.0*(2.0*h_av+y_lo-y_up)/
(y_up*y_up*y_up-y_lo*y_lo*y_lo-h_av*y_up*
y_up*y_up+h_av*y_lo*y_lo-y_lo-3.0*y_lo*y_up*y_up+
3.0*y_lo*y_lo*y_up+3.0*y_lo*y_up*y_up*h_av-3.0*y_lo*y_lo*y_up*h_av);

// y coordinate
double y=x[1];

// Parallel, parabolic inflow
veloc[0]=-1.0+C*(1.0-h_av)*((y_lo-y)*(y_up-y));
veloc[1]=0.0;
}

}

//=====
// "Bretherton element" is a fluid element (of type ELEMENT) for which
// the (inflow) velocity at those nodes that are located on a
// specified Mesh boundary is prescribed by Dirichlet boundary
// conditions. The key is that prescribed velocity profile can be
// a function of some external Data -- this dependency must
// be taken into account when computing the element's Jacobian
// matrix.
//
// This element type is useful, for instance, in the Bretherton
// problem, where the parabolic "inflow" profile is a function
// of the film thickness (represented by Spine heights) at the
// "outflow".
//=====
template<class ELEMENT>
class BrethertonElement : public ELEMENT
{
public:

// Typedef for pointer (global) function that specifies the
// the inflow
typedef void (*InflowFctPt)(const Vector<double>& x, Vector<double>& veloc);

// Constructor: Call the constructor of the underlying element
BrethertonElement() : ELEMENT() {}

// Activate the dependency of the "inflow" on the external
// data. Pass the vector of pointers to the external Data that affects
// the inflow, the id of the boundary on which the inflow
// condition is to be applied and the function pointer to
// to the global function that defines the inflow
void activate_inflow_dependency_on_external_data(
const Vector<Data*>& inflow_ext_data,
const unsigned& inflow_boundary,
InflowFctPt inflow_fct_pt)
{
// Copy data that affects the inflow
unsigned n_ext=inflow_ext_data.size();
Inflow_ext_data.resize(n_ext);
for (unsigned i=0;i<n_ext;i++)
{
Inflow_ext_data[i]=inflow_ext_data[i];
}
// Set inflow boundary
Inflow_boundary=inflow_boundary;
// Set fct pointer to inflow condition
Inflow_fct_pt=inflow_fct_pt;
}

// Overload assign local equation numbers: Add the dependency
// on the external Data that affects the inflow profile
void assign_local_eqn_numbers(const bool &store_local_dof_pt)
{
// Call the element's local equation numbering procedure first
ELEMENT::assign_local_eqn_numbers(store_local_dof_pt);

// Now add the equation numbers for the Data that affects the inflow
// profile

// Number of local equations so far
unsigned local_eqn_count = this->ndof();

// Find out max. number of values stored at all Data values

```

```

// that affect the inflow
unsigned max_nvalue=0;
unsigned n_ext=Inflow_ext_data.size();
for (unsigned i=0;i<n_ext;i++)
{
    // The external Data:
    Data* data_pt=Inflow_ext_data[i];
    // Number of values
    unsigned n_val=data_pt->nvalue();
    if (n_val>max_nvalue) max_nvalue=n_val;
}

// Allocate sufficient storage
Inflow_ext_data_eqn.resize(n_ext,max_nvalue);

//A local queue to store the global equation numbers
std::deque<unsigned long> global_eqn_number_queue;

// Loop over external Data that affect the "inflow"
for (unsigned i=0;i<n_ext;i++)
{
    // The external Data:
    Data* data_pt=Inflow_ext_data[i];

    // Loop over number of values:
    unsigned n_val=data_pt->nvalue();
    for (unsigned ival=0;ival<n_val;ival++)
    {
        // Is it free or pinned?
        long eqn_number=data_pt->eqn_number(ival);
        if (eqn_number>=0)
        {
            // Add global equation number to local storage
            global_eqn_number_queue.push_back(eqn_number);
            // Store local equation number associated with this external dof
            Inflow_ext_data_eqn(i,ival)=local_eqn_count;
            // Increment counter for local dofs
            local_eqn_count++;
        }
        else
        {
            Inflow_ext_data_eqn(i,ival)=-1;
        }
    }
}

//Now add our global equations numbers to the internal element storage
this->add_global_eqn_numbers(global_eqn_number_queue,
                             GeneralisedElement::Dof_pt_deque);
}

/// Overloaded Jacobian computation: Computes the Jacobian
/// of the underlying element and then adds the FD
/// operations to evaluate the derivatives w.r.t. the Data values
/// that affect the inflow.
void get_jacobian(Vector<double>& residuals,
                  DenseMatrix<double>& jacobian)
{
    // Loop over Data values that affect the inflow
    unsigned n_ext=Inflow_ext_data.size();

    // Call "normal" jacobian first.
    ELEMENT::get_jacobian(residuals, jacobian);

    if (n_ext==0) return;

    // Get ready for FD operations
    Vector<double> residuals_plus(residuals.size());
    double fd_step=1.0e-8;

    // Loop over Data values that affect the inflow
    for (unsigned i=0;i<n_ext;i++)
    {
        // Get Data item
        Data* data_pt=Inflow_ext_data[i];

        // Loop over values
        unsigned n_val=data_pt->nvalue();
        for (unsigned ival=0;ival<n_val;ival++)
        {
            // Local equation number
            int local_eqn=Inflow_ext_data_eqn(i,ival);

            // Dof or pinned?
            if (local_eqn>=0)

```

```

{
    //get pointer to the data value
    double *value_pt = data_pt->value_pt(ival);

    //Backup Data value
    double backup = *value_pt;

    // Do FD step
    *value_pt += fd_step;

    // Re-assign the inflow velocities for nodes in this element
    reassign_inflow();

    // Fill in the relevant column in the Jacobian matrix
    unsigned n_dof = this->ndof();
    //Zero the residuals
    for(unsigned idof=0;idof<n_dof;idof++) {residuals_plus[idof] = 0.0;}
    // Re-compute the element residual
    this->get_residuals(residuals_plus);

    for(unsigned idof=0;idof<n_dof;idof++)
    {
        jacobian(idof,local_eqn)=
            (residuals_plus[idof]-residuals[idof])/fd_step;
    }

    //Reset spine height
    *value_pt = backup;
}
// Note: Re-assignment of inflow is done on the fly during next loop
}
}

// Final re-assign for the inflow velocities for nodes in this element
reassign_inflow();
}
private:

/// For all nodes that are located on specified boundary
/// re-assign the inflow velocity, using the function pointed to
/// by the function pointer
void reassign_inflow()
{
    // Loop over all nodes in element -- if they are
    // on inflow boundary, re-assign their velocities
    Vector<double> x(2);
    Vector<double> veloc(2);
    unsigned n_nod = this->nnode();
    for (unsigned j=0;j<n_nod;j++)
    {
        Node* nod_pt = this->node_pt(j);

        if(nod_pt->is_on_boundary(Inflow_boundary))
        {
#ifdef PARANOID
            for (unsigned i=0;i<2;i++)
            {
                if (nod_pt->eqn_number(0)>=0)
                {
                    std::ostringstream error_stream;
                    error_stream
                        << "We're assigning a Dirichlet condition for the "
                        << i << "-th "
                        << "velocity, even though it is not pinned!\n"
                        << "This can't be right! I'm bailing out..."
                        << std::endl;

                    throw OomphLibError(error_stream.str(),
                                         OOMPH_CURRENT_FUNCTION,
                                         OOMPH_EXCEPTION_LOCATION);
                }
            }
#endif
            // Get inflow profile
            x[0]=nod_pt->x(0);
            x[1]=nod_pt->x(1);
            Inflow_fct_pt(x,veloc);
            nod_pt->set_value(0,veloc[0]);
            nod_pt->set_value(1,veloc[1]);
        }
    }
}
}

```

```

/// Storage for the external Data that affects the inflow
Vector<Data*> Inflow_ext_data;

/// Storage for the local equation numbers associated the Data
/// values that affect the inflow
DenseMatrix<int> Inflow_ext_data_eqn;

/// Number of the inflow boundary in the global mesh
unsigned Inflow_boundary;

/// Function pointer to the global function that specifies the
/// inflow velocity profile on the global mesh boundary Inflow_boundary
InflowFctPt Inflow_fct_pt;

};

// Note: Specialisation must go into namespace!
namespace oomph
{

//=====
/// Face geometry of the Bretherton 2D Crouzeix_Raviart spine elements
//=====
template<>
class FaceGeometry<BrethertonElement<SpineElement<QCrouzeixRaviartElement<2> > > >: public virtual
    QElement<1,3>
{
public:
    FaceGeometry() : QElement<1,3>() {}
};

//=====
/// Face geometry of the Bretherton 2D Taylor Hood spine elements
//=====
template<>
class FaceGeometry<BrethertonElement<SpineElement<QTaylorHoodElement<2> > > >: public virtual QElement<1,3>
{
public:
    FaceGeometry() : QElement<1,3>() {}
};

//=====
/// Face geometry of the Face geometry of the
/// the Bretherton 2D Crouzeix_Raviart spine elements
//=====
template<>
class FaceGeometry<FaceGeometry<BrethertonElement<
    SpineElement<QCrouzeixRaviartElement<2> > > >: public virtual PointElement
{
public:
    FaceGeometry() : PointElement() {}
};

//=====
/// Face geometry of face geometry of
/// the Bretherton 2D Taylor Hood spine elements
//=====
template<>
class FaceGeometry<FaceGeometry<BrethertonElement<SpineElement<QTaylorHoodElement<2> > > >: public virtual
    PointElement
{
public:
    FaceGeometry() : PointElement() {}
};

}

////////////////////////////////////
////////////////////////////////////
////////////////////////////////////

//=====
/// Bretherton problem
//=====
template<class ELEMENT>
class BrethertonProblem : public Problem

```

```

{

public:

    /// Constructor:
    BrethertonProblem();

    /// Spine heights/lengths are unknowns in the problem so their
    /// values get corrected during each Newton step. However,
    /// changing their value does not automatically change the
    /// nodal positions, so we need to update all of them
    void actions_before_newton_convergence_check()
    {
        // Update
        Bulk_mesh_pt->node_update();

        // Apply inflow on boundary 1
        unsigned ibound=1;
        unsigned num_nod=mesh_pt()->nboundary_node(ibound);

        // Coordinate and velocity
        Vector<double> x(2);
        Vector<double> veloc(2);

        // Loop over all nodes
        for (unsigned inod=0; inod<num_nod; inod++)
        {
            // Get nodal position
            x[0]=mesh_pt()->boundary_node_pt(ibound,inod)->x(0);
            x[1]=mesh_pt()->boundary_node_pt(ibound,inod)->x(1);

            // Get inflow profile
            Global_Physical_Variables::inflow(x, veloc);
            mesh_pt()->boundary_node_pt(ibound,inod)->set_value(0, veloc[0]);
            mesh_pt()->boundary_node_pt(ibound,inod)->set_value(1, veloc[1]);
        }
    }

    /// Update before solve: empty
    void actions_before_newton_solve() {}

    /// Update after solve can remain empty, because the update
    /// is performed automatically after every Newton step.
    void actions_after_newton_solve() {}

    /// Fix pressure value 1 in element e to value p_value
    void fix_pressure(const unsigned &e, const unsigned &l,
                     const double &pvalue)
    {
        //Fix the pressure at that element
        dynamic_cast<ELEMENT *>(Bulk_mesh_pt->element_pt(e))->
            fix_pressure(1,pvalue);
    }

    /// Activate the dependency of the inflow velocity on the
    /// spine heights at the outflow
    void activate_inflow_dependency();

    /// Run a parameter study; perform specified number of steps
    void parameter_study(const unsigned& nsteps);

    /// Doc the solution
    void doc_solution(DocInfo& doc_info);

private:

    /// Pointer to control element
    ELEMENT* Control_element_pt;

    /// Trace file
    ofstream Trace_file;

    /// Pointer to bulk mesh
    BrethertonSpineMesh<ELEMENT, SpineLineFluidInterfaceElement<ELEMENT> > *
    Bulk_mesh_pt;

};

//=====
/// Problem constructor
//=====
template<class ELEMENT>
BrethertonProblem<ELEMENT>::BrethertonProblem()

```

```

{

// Number of elements in the deposited film region
unsigned nx1=24;

// Number of elements in the bottom part of the transition region
unsigned nx2=6;

// Number of elements in channel region
unsigned nx3=12;

// Number of elements in the vertical part of the transition region
// (=half the number of elements in the liquid filled region ahead
// of the finger tip)
unsigned nhalf=4;

// Number of elements through thickness of deposited film
unsigned nh=3;
// Thickness of deposited film
double h=0.1;

// Create wall geom objects
GeomObject* lower_wall_pt=new StraightLine(-1.0);
GeomObject* upper_wall_pt=new StraightLine( 1.0);

// Start coordinate on wall
double xi0=-4.0;

// End of transition region on wall
double xi1=1.5;

// End of liquid filled region (inflow) on wall
double xi2=5.0;

//Now create the mesh
Bulk_mesh_pt = new BrethertonSpineMesh<ELEMENT,
    SpineLineFluidInterfaceElement<ELEMENT> >
    (nx1,nx2,nx3,nh,nhalf,h,lower_wall_pt,upper_wall_pt,xi0,0.0,xi1,xi2);

// Make bulk mesh the global mesh
mesh_pt()=Bulk_mesh_pt;

// Store the control element
Control_element_pt=Bulk_mesh_pt->control_element_pt();

// Set pointers to quantities that determine the inflow profile

// Film thickness at outflow on lower wall:
Global_Physical_Variables::H_lo_pt=
    Bulk_mesh_pt->spine_pt(0)->spine_height_pt()->value_pt(0);

// Film thickness at outflow on upper wall:
unsigned last_spine=Bulk_mesh_pt->nfree_surface_spines()-1;
Global_Physical_Variables::H_up_pt=
    Bulk_mesh_pt->spine_pt(last_spine)->spine_height_pt()->value_pt(0);

// Current y-position on lower wall at inflow
unsigned ibound=1;
Global_Physical_Variables::Y_lo_pt=
    Bulk_mesh_pt->boundary_node_pt(ibound,0)->x_pt(0,1);

// Current y-position on upper wall at inflow
unsigned nnod=Bulk_mesh_pt->nboundary_node(ibound);
Global_Physical_Variables::Y_up_pt=
    Bulk_mesh_pt->boundary_node_pt(ibound,nnod-1)->x_pt(0,1);

// Activate dependency of inflow on spine heights at outflow
activate_inflow_dependency();

// Set the boundary conditions for this problem: All nodes are
// free by default -- just pin the ones that have Dirichlet conditions
// here

// No slip on boundaries 0 1 and 2
for(unsigned ibound=0;ibound<=2;ibound++)
{
    unsigned num_nod=mesh_pt()->nboundary_node(ibound);
    for(unsigned inod=0;inod<num_nod;inod++)
    {
        mesh_pt()->boundary_node_pt(ibound,inod)->pin(0);
        mesh_pt()->boundary_node_pt(ibound,inod)->pin(1);
    }
}

// Uniform, parallel outflow on boundaries 3 and 5
for(unsigned ibound=3;ibound<=5;ibound+=2)

```

```

{
    unsigned num_nod=mesh_pt()->nboundary_node(ibound);
    for (unsigned inod=0; inod<num_nod; inod++)
    {
        mesh_pt()->boundary_node_pt(ibound, inod)->pin(0);
        mesh_pt()->boundary_node_pt(ibound, inod)->pin(1);
    }
}

// Pin central spine
unsigned central_spine=(Bulk_mesh_pt()->nfree_surface_spines()-1)/2;
Bulk_mesh_pt()->spine_pt(central_spine)->spine_height_pt()->pin(0);

// No slip in moving frame on boundaries 0 and 2
for (unsigned ibound=0; ibound<=2; ibound+=2)
{
    unsigned num_nod=mesh_pt()->nboundary_node(ibound);
    for (unsigned inod=0; inod<num_nod; inod++)
    {
        // Parallel flow
        mesh_pt()->boundary_node_pt(ibound, inod)->set_value(0, -1.0);
        mesh_pt()->boundary_node_pt(ibound, inod)->set_value(1, 0.0);
    }
}

// Parallel, uniform outflow on boundaries 3 and 5
for (unsigned ibound=3; ibound<=5; ibound+=2)
{
    unsigned num_nod=mesh_pt()->nboundary_node(ibound);
    for (unsigned inod=0; inod<num_nod; inod++)
    {
        // Parallel inflow
        mesh_pt()->boundary_node_pt(ibound, inod)->set_value(0, -1.0);
        mesh_pt()->boundary_node_pt(ibound, inod)->set_value(1, 0.0);
    }
}

//Complete the problem setup to make the elements fully functional

//Loop over the elements in the layer
unsigned n_bulk=Bulk_mesh_pt()->nbulk();
for(unsigned i=0; i<n_bulk; i++)
{
    //Cast to a fluid element
    ELEMENT *el_pt = dynamic_cast<ELEMENT*>(Bulk_mesh_pt()->
                                           bulk_element_pt(i));

    //Set the Reynolds number, etc
    el_pt->re_pt() = &Global_Physical_Variables::Re;
    el_pt->re_st_pt() = &Global_Physical_Variables::ReSt;
    el_pt->re_invfr_pt() = &Global_Physical_Variables::ReInvFr;
    el_pt->g_pt() = &Global_Physical_Variables::G;
}

//Loop over 1D interface elements and set capillary number
unsigned interface_element_pt_range = Bulk_mesh_pt()->ninterface_element();
for(unsigned i=0; i<interface_element_pt_range; i++)
{
    //Cast to a interface element
    SpineLineFluidInterfaceElement<ELEMENT*> el_pt =
        dynamic_cast<SpineLineFluidInterfaceElement<ELEMENT*>*>
        (Bulk_mesh_pt()->interface_element_pt(i));

    //Set the Capillary number
    el_pt->ca_pt() = &Global_Physical_Variables::Ca;
}

//Update the nodal positions, so that the domain is correct
//(and therefore the repeated node test passes)
Bulk_mesh_pt()->node_update();

//Do equation numbering
cout << "Number of unknowns: " << assign_eqn_numbers() << std::endl;
}

//=====
/// Activate the dependency of the inflow velocity on the
/// spine heights at the outflow
//=====
template<class ELEMENT>
void BrethertonProblem<ELEMENT>::activate_inflow_dependency()

```

```

{
    // Spine heights that affect the inflow
    Vector<Data*> outflow_spines(2);

    // First spine
    outflow_spines[0]=Bulk_mesh_pt->spine_pt(0)->spine_height_pt();

    // Last proper spine
    unsigned last_spine=Bulk_mesh_pt->nfree_surface_spines()-1;
    outflow_spines[1]=Bulk_mesh_pt->spine_pt(last_spine)->spine_height_pt();

    /// Loop over elements on inflow boundary (1)
    unsigned ibound=1;
    unsigned nel=Bulk_mesh_pt->nboundary_element(ibound);
    for (unsigned e=0;e<nel;e++)
    {
        // Get pointer to element
        ELEMENT* el_pt=dynamic_cast<ELEMENT*>(Bulk_mesh_pt->
                                              boundary_element_pt(ibound,e));

        // Activate dependency on inflow
        el_pt->activate_inflow_dependency_on_external_data(
            outflow_spines,ibound,&Global_Physical_Variables::inflow);
    }
}

//=====
/// Doc the solution
//=====
template<class ELEMENT>
void BrethertonProblem<ELEMENT>::doc_solution(DocInfo& doc_info)
{
    ofstream some_file;
    char filename[100];

    // Number of plot points
    unsigned npts=5;

    // Control coordinate: At bubble tip
    Vector<double> s(2);
    s[0]=1.0;
    s[1]=1.0;
    // Last proper spine
    unsigned last_spine=Bulk_mesh_pt->nfree_surface_spines()-1;

    // Doc
    Trace_file << " " << Global_Physical_Variables::Ca;
    Trace_file << " " << Bulk_mesh_pt->spine_pt(0)->height();
    Trace_file << " " << Bulk_mesh_pt->spine_pt(last_spine)->height();
    Trace_file << " " << 1.3375*pow(Global_Physical_Variables::Ca,2.0/3.0);
    Trace_file << " " << -Control_element_pt->interpolated_p_nst(s)*
        Global_Physical_Variables::Ca;
    Trace_file << " " << 1.0+3.8*pow(Global_Physical_Variables::Ca,2.0/3.0);
    Trace_file << " " << Control_element_pt->interpolated_u_nst(s,0);
    Trace_file << " " << Control_element_pt->interpolated_u_nst(s,1);
    Trace_file << std::endl;

    // Output solution
    snprintf(filename, sizeof(filename), "%s/soln%i.dat",doc_info.directory().c_str(),
        doc_info.number());
    some_file.open(filename);
    Bulk_mesh_pt->output(some_file,npts);
    some_file.close();

    // Output boundaries
    snprintf(filename, sizeof(filename), "%s/boundaries%i.dat",doc_info.directory().c_str(),
        doc_info.number());
    some_file.open(filename);
    Bulk_mesh_pt->output_boundaries(some_file);
    some_file.close();
}

//=====
/// Parameter study
//=====

```

```

template<class ELEMENT>
void BrethertonProblem<ELEMENT>::parameter_study(const unsigned& nsteps)
{
    // Increase maximum residual
    Problem::Max_residuals=500.0;

    // Set output directory
    DocInfo doc_info;
    doc_info.set_directory("RESLT");
    doc_info.number()=0;

    // Open trace file
    char filename[100];
    snprintf(filename, sizeof(filename), "%s/trace.dat", doc_info.directory().c_str());
    Trace_file.open(filename);

    Trace_file << "VARIABLES=\"Ca\", ";
    Trace_file << "\"h<sub>bottom</sub>\", \"h<sub>top</sub>\", ";
    Trace_file << "\"h<sub>Bretherton</sub>\", \"p<sub>tip</sub>\", ";
    Trace_file << "\"p<sub>tip (Bretherton)</sub>\", \"u<sub>stag</sub>\", ";
    Trace_file << "\"v<sub>stag</sub>\", ";
    Trace_file << "\"<greek>a</greek><sub>bottom</sub>\", ";
    Trace_file << "\"<greek>a</greek><sub>top</sub>\", ";
    Trace_file << std::endl;

    // Initial scaling factor for Ca reduction
    double factor=2.0;

    //Doc initial solution
    doc_solution(doc_info);

    //Loop over the steps
    for (unsigned step=1; step<=nsteps; step++)
    {
        cout << std::endl << "STEP " << step << std::endl;

        // Newton method tends to converge if initial stays bounded below
        // a certain tolerance: This is cheaper to check than restarting
        // the Newton method after divergence (we'd also need to back up
        // the previous solution)
        double maxres = Problem::Max_residuals;
        while (true)
        {
            cout << "Checking max. res for Ca = "
                 << Global_Physical_Variables::Ca << ".    Max. residual: ";

            // Check the maximum residual
            DoubleVector residuals;
            actions_before_newton_solve();
            actions_before_newton_convergence_check();
            get_residuals(residuals);
            double max_res=residuals.max();
            cout << max_res;

            // Check what to do
            if (max_res>maxres)
            {
                cout << ". Too big!" << std::endl;
                // Reset the capillary number
                Global_Physical_Variables::Ca *= factor;
                // Reduce the factor
                factor=1.0+(factor-1.0)/1.5;
                cout << "New reduction factor: " << factor << std::endl;
                // Reduce the capillary number
                Global_Physical_Variables::Ca /= factor;
                // Try again
                continue;
            }
            // Looks promising: Let's proceed to solve
            else
            {
                cout << ". OK" << std::endl << std::endl;
                // Break out of the Ca adjustment loop
                break;
            }
        }

        //Solve
        cout << "Solving for capillary number: "
             << Global_Physical_Variables::Ca << std::endl;
        newton_solve();

        // Doc solution
        doc_info.number()++;
        doc_solution(doc_info);
    }
}

```

```

    // Reduce the capillary number
    Global_Physical_Variables::Ca /= factor;
}

}

//=====
/// Driver code for unsteady two-layer fluid problem. If there are
/// any command line arguments, we regard this as a validation run
/// and perform only a single step.
//=====
int main(int argc, char* argv[])
{

    // Store command line arguments
    CommandLineArgs::setup(argc,argv);

    // Set physical parameters:

    //Set direction of gravity: Vertically downwards
    Global_Physical_Variables::G[0] = 0.0;
    Global_Physical_Variables::G[1] = -1.0;

    // Womersley number = Reynolds number (St = 1)
    Global_Physical_Variables::ReSt = 0.0;
    Global_Physical_Variables::Re = Global_Physical_Variables::ReSt;

    // The Capillary number
    Global_Physical_Variables::Ca = 0.05;

    // Re/Fr -- a measure of gravity...
    Global_Physical_Variables::ReInvFr = 0.0;

    //Set up the problem
    BrethertonProblem<BrethertonElement<SpineElement<QCrouzeixRaviartElement<2> > > > problem;

    // Self test:
    problem.self_test();

    // Number of steps:
    unsigned nstep;
    if (CommandLineArgs::Argc>1)
    {
        // Validation run: Just one step
        nstep=2;
    }
    else
    {
        // Full run otherwise
        nstep=30;
    }

    // Run the parameter study: Perform nstep steps
    problem.parameter_study(nstep);

}

```

## 1.5 Source files for this tutorial

- The source files for this tutorial are located in the directory:

`demo_drivers/navier_stokes/bretherton`

- The driver code is:

`demo_drivers/navier_stokes/bretherton/bretherton.cc`

## 1.6 PDF file

A [pdf version](#) of this document is available. \